Zoho Analytics Integration

1. Overview

1.1 Description

Zoho Analytics Integration connects **Zoho Analytics** (business intelligence and reporting platform) with **Newo Platform**.

It enables:

- Scope-matching the caller against the Vendors directory by company name (no phone-based identity check — see §1.4)
- Looking up vendor invoice (Bill) status by load number + vendor (due date, paid status, amount due, payee)
- Verifying rate details by load number (rate / amount, payee)

This integration is designed for **logistics companies running QuickBooks Desktop** who need **an AI phone agent to answer vendor (carrier/transporter) inquiries about invoices and rates** using data exported from QuickBooks into Zoho Analytics tables.

1.2 Use Cases

- When a vendor calls and asks about an invoice → Agent verifies identity, queries Bills (filtered by load number in `Memo` and the verified vendor's `Vendor ID`), and reads back due date, paid status, amount due, and payee
- When a vendor calls to verify a rate → Agent queries Bills by load number, then resolves the payee from Vendors, and reads back rate (= bill amount), payee, and payment amount
- When a caller needs to be scope-matched → Agent searches Vendors by company name (LIKE matching); only **active** vendors are accepted. Phone number is intentionally NOT used — see §1.4.

1.4 A note on caller "verification"

The integration's caller-verification step is a **scope filter**, not authentication:

- Caller ID can be spoofed at the SIP level — using `Phone` as a verification factor offers near-zero security.
- In QBD, `Vendor.Phone` is typically the office contact, while real calls usually come from drivers / dispatchers on personal mobiles. Exact-string matching reliably produces false negatives.
- Phone format in QBD is unnormalized (`(555) 123-4567` , `+15551234567` , `5551234567` …); E.164-normalized caller numbers do not match without per-record normalization that the integration does not perform.

Consequently, verification is intentionally limited to `Vendor Name LIKE '%name%' AND Is Active='Yes'` . The result is enough to scope a query ("read out **this** vendor's bills") without claiming to authenticate the caller. If you later need stronger identity proof, prefer a knowledge-factor challenge (e.g. *"name a recent load number from your account"*) over phone matching.

1.3 Supported Features

Feature	Supported	Notes
Invoice Status Lookup	✅	Bills filtered by `Memo` (load number) + `Vendor ID`

Rate Verification	✅	Bills by `Memo`; rate = `Amount Due`, payee = Vendor Name
Caller Scope-Match	✅	Vendors `LIKE %name%` + `Is Active='Yes'`; **not** authentication (see §1.4)
Multi-Region Support	✅	US, EU, IN, AU, JP, SA, CA data centers
OAuth2 Authentication	✅	Standard redirect flow with token auto-refresh
Custom Canvas Scenarios	✅	Logistics-specific scenarios (Invoice Status, Rate Verification)
Booking/Scheduling	❌	Not applicable — this is a data query integration
Write Operations	❌	Read-only — agent never modifies Zoho Analytics data

2. Requirements

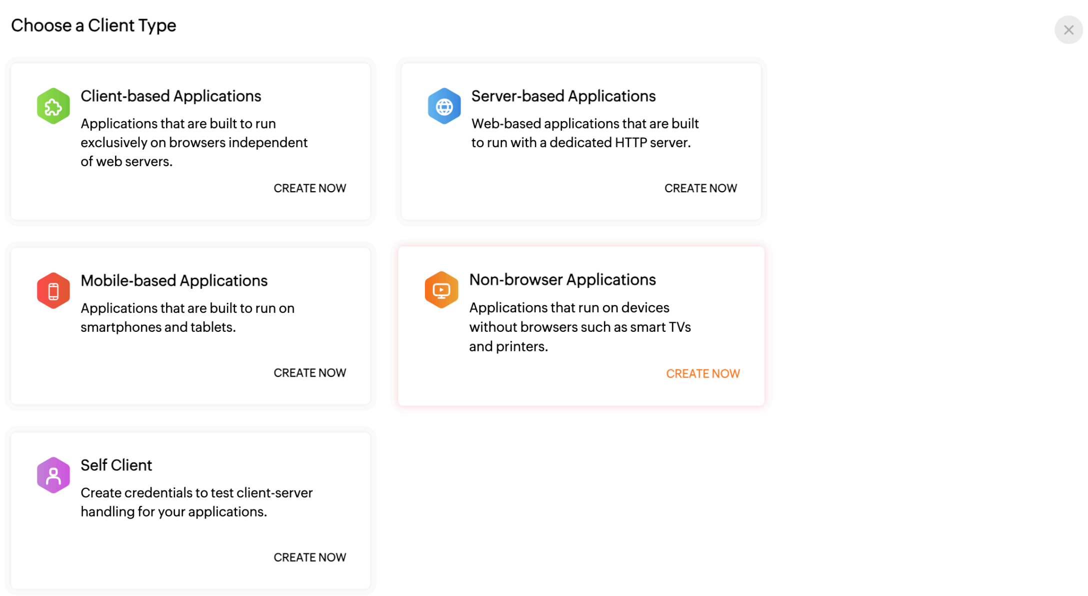
Before installation:

- Active **Zoho Analytics** account with API access (free tier works)
- **Server-based Applications** client registered in [Zoho API Console]
- A Zoho Analytics workspace with two tables (typical QuickBooks Desktop export): **Bills** and **Vendors**
- Load numbers must be stored in **`Bills.Memo`** (this is how QuickBooks Desktop typically captures them)

3. How to Setup

3.1 Step 1 — Register OAuth Client in Zoho

1. Go to [Zoho API Console]
2. Click **Add Client** → select **Server-based Applications**
3. Fill in:
    - **Client Name**: `Newo Zoho Analytics`
    - **Homepage URL**: `https://newo.ai`
    - **Authorized Redirect URIs**: `https://static.newo.ai/oauth/index.html`
4. Copy **Client ID** and **Client Secret**



3.2 Step 2 — Prepare Tables in Zoho Analytics

The integration reads from two tables. Column names must match exactly (case-sensitive, with spaces).

Bills — vendor invoices (Accounts Payable). Load number lives in `Memo` .

Column Name	Data Type	Used by integration
Bill ID	Plain Text	join key
Vendor ID	Plain Text	join key to Vendors
Bill Date	Date	optional readback
Due Date	Date	readback
Amount Due	Number	rate / amount due
Is Paid	Plain Text	readback
Paid Status	Plain Text	readback
Memo	Plain Text	**load number** (LIKE)

Vendors — vendor (carrier/transporter) directory.

Column Name	Data Type	Used by integration
Vendor ID	Plain Text	join key from Bills

Vendor Name	Plain Text	caller verification (LIKE) + payee name
Is Active	Plain Text	only `Yes` matches verification
Phone	Plain Text	optional read-back to the agent (NOT a verification factor — see §1.4)
Email	Plain Text	optional readback
Contact	Plain Text	optional readback
Company Name	Plain Text	optional readback

> ***Important***: *Column names must match exactly (case-sensitive, with spaces). The integration queries by these exact column names.*

3.3 Step 3 — Get Workspace ID and View IDs

Open each table in Zoho Analytics. The View ID is visible in the browser URL:

```
https://analytics.zoho.eu/workspace/WORKSPACE_ID/view/VIEW_ID
```

Note down:

- **Workspace ID**
- **View ID** for the **Bills** table
- **View ID** for the **Vendors** table

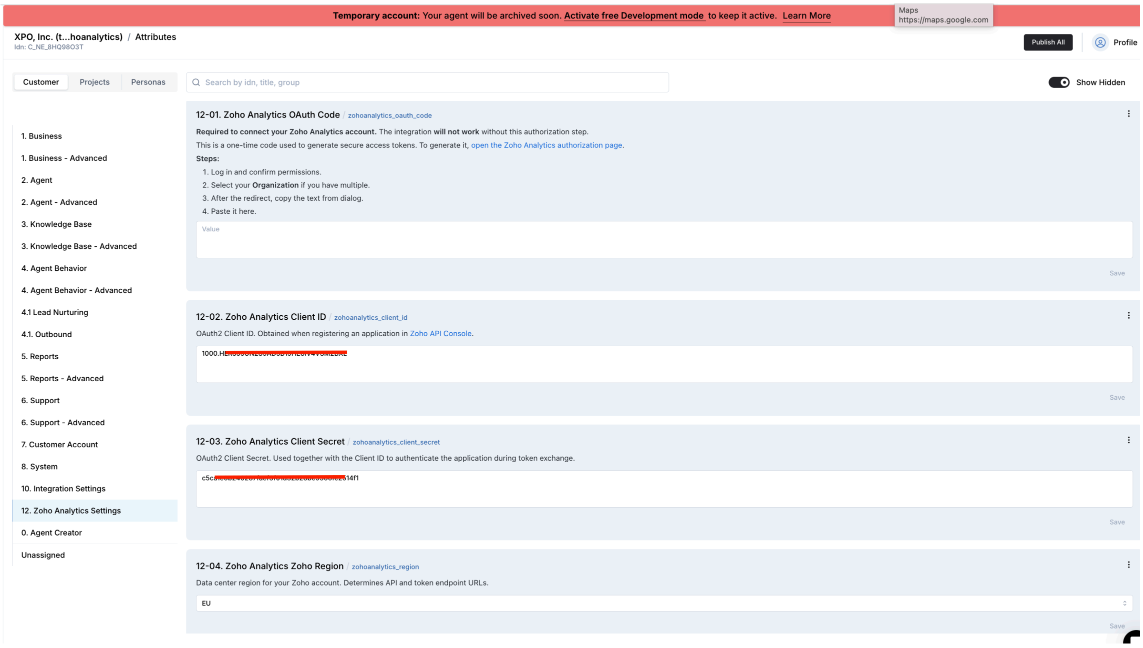
The integration's UI will populate dropdowns for both views automatically once the OAuth flow completes — you can pick them visually and the IDs will be derived for you.

3.4 Step 4 — Connect in Newo Platform

1. Open **builder.newo.ai → Project → Attributes**
2. Set the following attributes (visible to operator):

Attribute	Required	Description
`01. OAuth Code`	✅	Paste the code from the Zoho redirect (one-time)
`02. Client ID`	✅	OAuth2 Client ID from Zoho API Console
`03. Client Secret`	✅	OAuth2 Client Secret from Zoho API Console
`04. Zoho Region`	✅	Data center: US, EU, IN, AU, JP, SA, or CA
`05. Organization`	✅	Pick the org from the auto-populated dropdown
`06. Workspace`	✅	Pick the workspace from the auto-populated dropdown
`07. Bills View`	✅	Pick the **Bills** table from the auto-populated dropdown

`08. Vendors View`	✅	Pick the **Vendors** table from the auto-populated dropdown
`Caller Verification Feature`	❌	Toggle for the verify-caller tool (default: `True`)
`Invoice Status Feature`	❌	Toggle for the invoice lookup tool (default: `True`)
`Rate Verification Feature`	❌	Toggle for the rate lookup tool (default: `True`)
`Automatically Add Default Scenarios to Canvas`	❌	Auto-publish canvas scenarios (default: `True`)



3. **Setup is 3-phase the first time**:
    - Phase 1 — Fill in OAuth Code, Client ID, Client Secret, Region. **Publish**.
    - Phase 2 — Pick the Organization from the dropdown that appeared. **Publish**.
    - Phase 3 — Pick the Workspace, then the Bills View and Vendors View. **Publish** once more.

After the third publish, the integration:

- Connects to the right Zoho data center based on your region
- Authorizes against Zoho with the OAuth code you provided
- Enables the three logistics tools (verify caller, invoice lookup, rate verification) for the agent
- Adds the matching scenarios to the Workflow Builder Canvas so the agent knows when to use each tool

4. How to Use

4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a code-phrase the agent says, detected by the ConvoAgent tool caller.

- An **Action** is a Zoho Analytics Sync Export query (the integration runs short read-only queries on the Bills and Vendors tables).

Available Triggers:

Trigger	Tool Event	Code-phrase
Verify Caller	`zohoanalytics_verify_caller_tool`	Agent says *"Let me verify your information..."*
Invoice Status Lookup	`zohoanalytics_query_invoice_status_tool`	Agent says *"Let me look up the invoice..."*
Rate Verification	`zohoanalytics_query_rate_verification_tool`	Agent says *"Let me check the rate..."*

What the agent does behind each trigger:

- **Verify Caller** — searches `Vendors` by `Vendor Name LIKE '%<name>%' AND Is Active='Yes'` . **Phone is intentionally not part of the filter** (see §1.4). On success, the matched `Vendor ID` is cached on the caller's persona so subsequent invoice / rate lookups skip a round-trip to the Vendors table.

- **Invoice Status Lookup** — runs in two sync steps to combine vendor identity with the bill row:

    1. **Vendors lookup** (skipped if the caller is already verified) — finds the `Vendor ID` from the caller-supplied vendor name.
    2. **Bills lookup** — pulls the bill row matching `Memo LIKE '%<load_number>%' AND Vendor ID='<id>'` and reads back due date, paid status, amount due, payee.

- **Rate Verification** — also two sync steps but in the reverse order:

    1. **Bills lookup** — finds the bill matching `Memo LIKE '%<load_number>%'` (no vendor filter — the caller does not need to be verified).
    2. **Vendors lookup** — resolves the payee name from `Vendor ID` , then reads back rate (= `Amount Due`), payee, and payment amount.

The same caller is never asked the same question twice in a single call — once a result is on screen, the agent answers from it without re-running the tool. Cached values are cleared automatically when the conversation ends.

4.2 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel) by interacting with the agent as a vendor calling about invoices or rates.

4.3 Testing and Verification

To test the integration:

1. **Setup**: Publish the project through all 3 phases and verify the dropdowns (Organization, Workspace, Bills View, Vendors View) populate and stay selected.
2. **OAuth**: Click the authorization link in the OAuth Code attribute, authorize in Zoho, paste the code back into the attribute, and publish again. The integration stores the tokens automatically.
3. **Caller Verification**: Say *"Hi, this is [vendor name from Vendors table]"* → Agent should say *"Let me verify your information"* → Confirm verification succeeded.

4. **Invoice Lookup**: Ask *"What's the status of my invoice for load [Memo value from Bills]?"* → Agent should return due date, paid status, amount due, payee.
5. **Rate Verification**: Ask *"What's the rate for load [Memo value]?"* → Agent should return rate, payee, payment amount.

If no action occurs:

- Ensure Client ID and Client Secret are set before the first Publish.
- Verify Region matches your Zoho account data center (check URL: `.zoho.com` = US, `.zoho.eu` = EU, etc.).
- Confirm both view dropdowns (Bills, Vendors) are picked — leaving the placeholder selected silently disables the feature.
- Re-publish after any attribute changes.
- Check that table column names match exactly: `Bill ID`, `Vendor ID`, `Memo`, `Due Date`, `Amount Due`, `Is Paid`, `Paid Status`, `Vendor Name`, `Is Active`, `Phone`, `Email`, `Contact`, `Company Name`.
- Verify data exists in Zoho Analytics tables (empty tables return "not found").
- Make sure load numbers in `Bills.Memo` are stored as bare identifiers (`LD34436`) — the lookup uses substring matching, so verbose memos like *"LD34436 — partial pay"* still match, but inconsistent prefixes can produce unexpected hits.

> *Note: This integration performs **read-only** queries against Zoho Analytics. No data is modified in Zoho Analytics through the agent.*

5. FAQ

Q: Does the integration modify data in Zoho Analytics? A: No. The integration is read-only — it only looks up rows. No data is created, updated, or deleted.

Q: How does data get into Zoho Analytics tables? A: Data must be synced from QuickBooks Desktop (or another source) to Zoho Analytics separately. Zoho Analytics supports imports from databases, CSV, cloud storage, and 250+ integrations. Data freshness depends on your sync frequency.

Q: What if a vendor's invoice has the load number outside the `Memo` column? A: The integration always reads from `Bills.Memo`. If your QuickBooks export stores the load number in a different field, copy or compute it into `Memo` during the sync to Zoho Analytics — that is the cheapest place to fix it.

Q: What happens if the vendor is not found during verification? A: The agent informs the caller that verification failed and offers to transfer to a human representative.

Q: Can the agent handle multiple invoice lookups in one call? A: Yes. After answering an invoice question, the agent asks *"Is there anything else?"* The caller can ask about another load number, and the second lookup reuses the already-verified vendor (no second Vendors query is performed).

Q: What if the Bills table has more than 1 million rows? A: The current setup is designed for tables up to 1 million rows. If you expect more, contact us — a different lookup approach will be needed.

Q: Which Zoho regions are supported? A: US, EU, India, Australia, Japan, Saudi Arabia, and Canada. Set the correct region in the **Region** attribute.

Q: Can I use the Zoho Analytics free tier? A: Yes. The free tier (2 users, 5 workspaces, 10,000 rows) is sufficient for testing and small deployments.